Building Interoperable Location-Based Augmented Reality with the Open AR Cloud

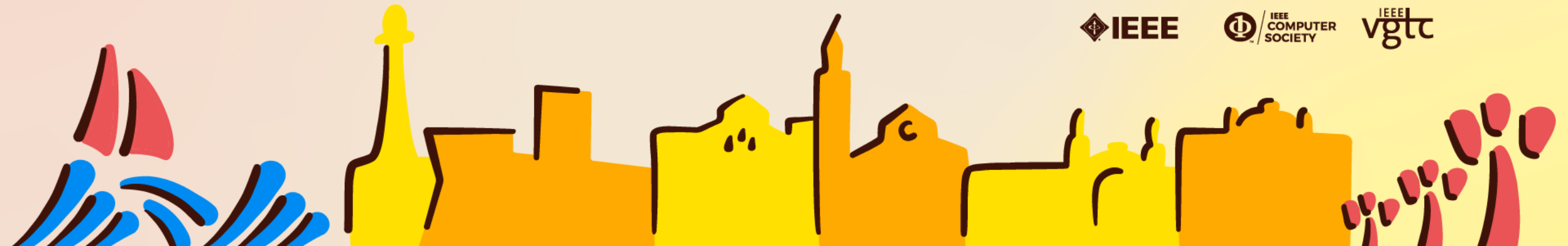
James Jackson, Open AR Cloud

ISMAR 2026

October 5 - 9, Bari, Italy

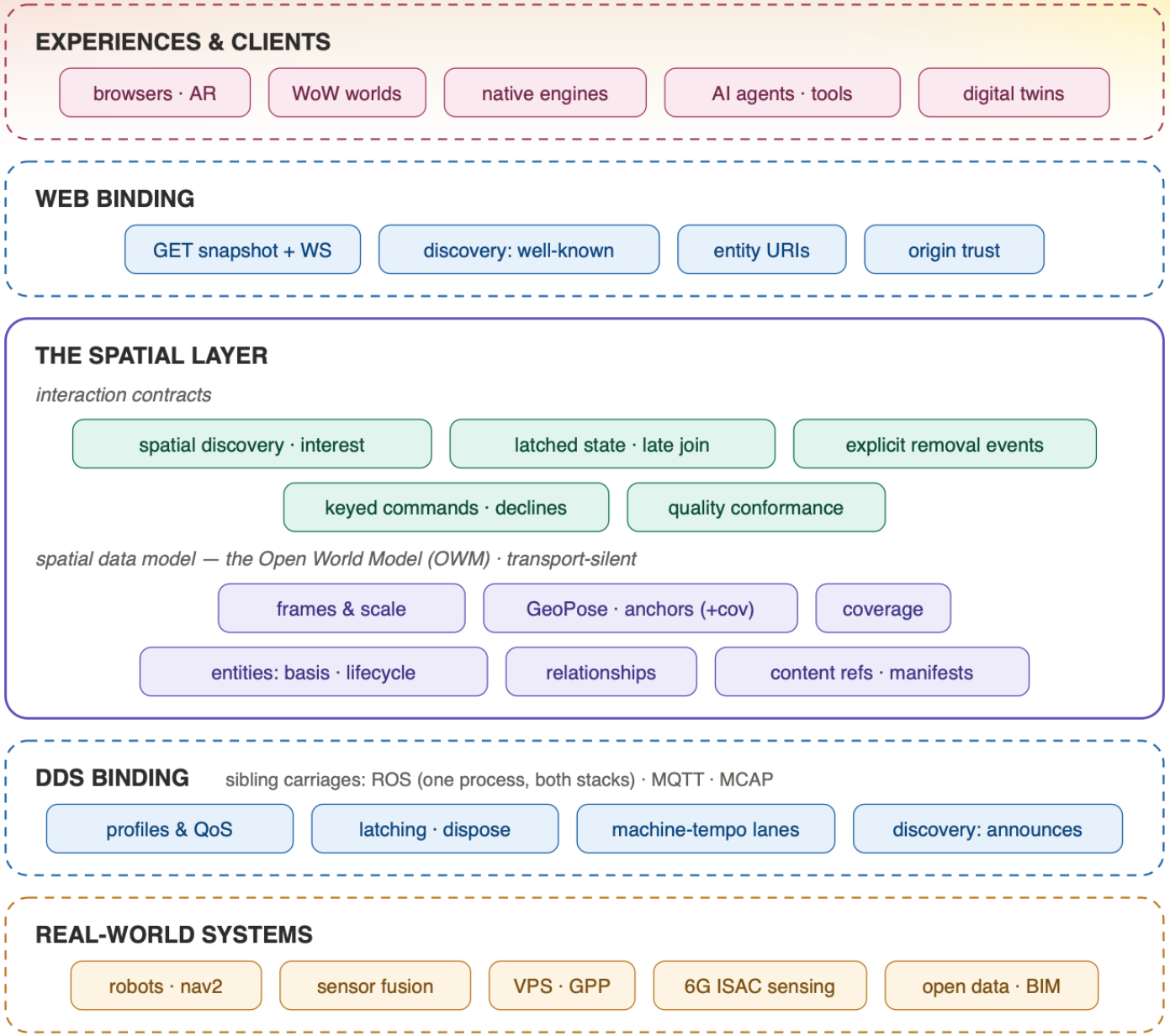
The 25th International Symposium on Mixed and Augmented Reality

IEEE IEEE COMPUTER SOCIETY IEEE VGTC



The missing spatial layer

Bridging AR, robotics, telecom (network-as-a-sensor), and digital twins



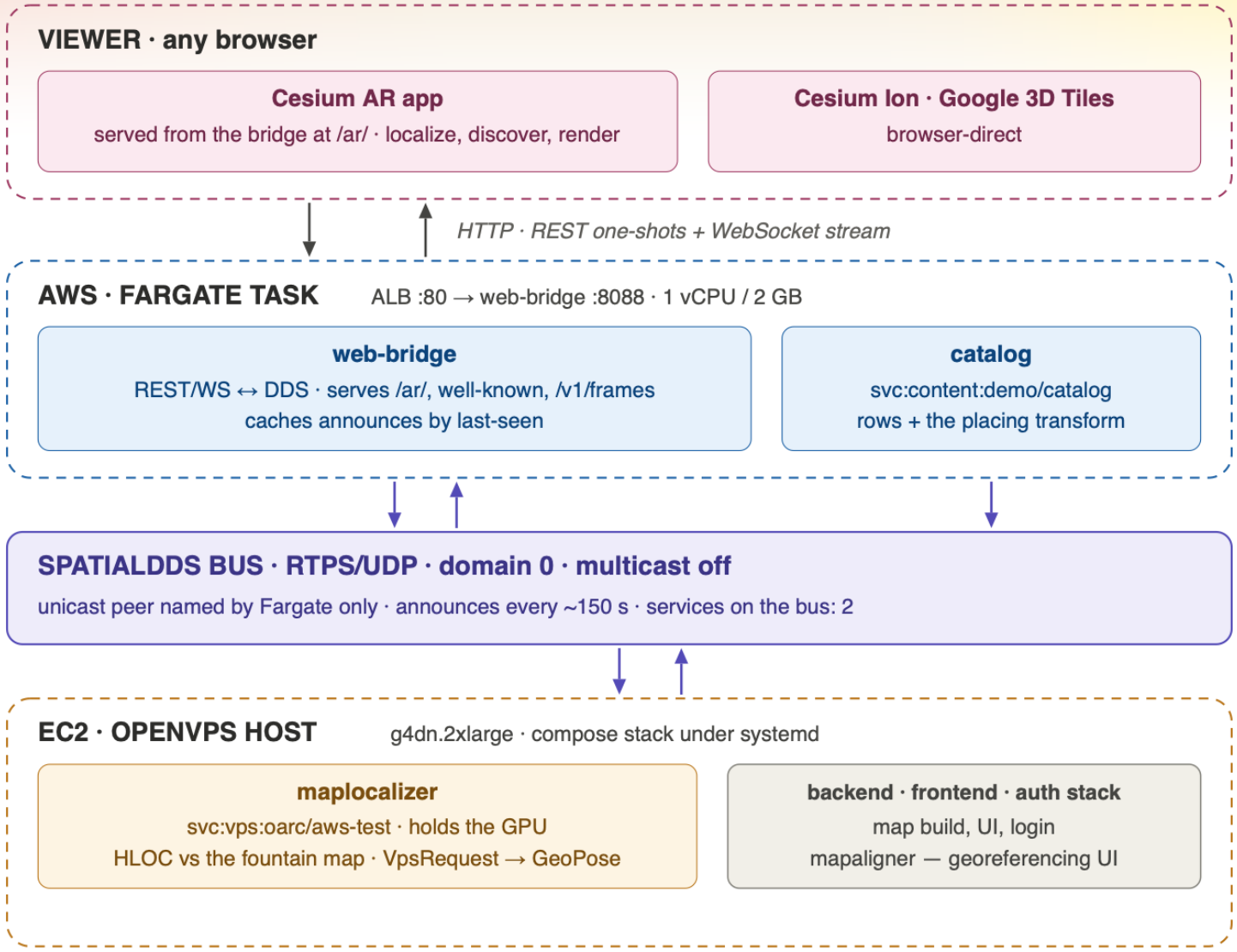
https://spatialdds.org

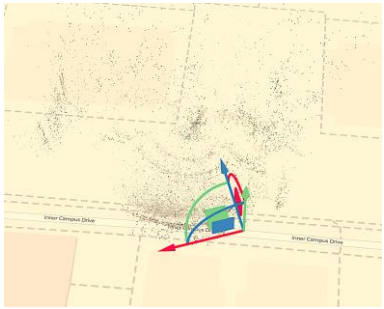
Validated against public datasets

LaMAR (multi-device AR): 70 tests
DeepSense 6G (V2I radar): 44
S3E (multi-robot SLAM): 38
ScanNet (indoor semantics): 35
nuScenes (automotive AV): 27

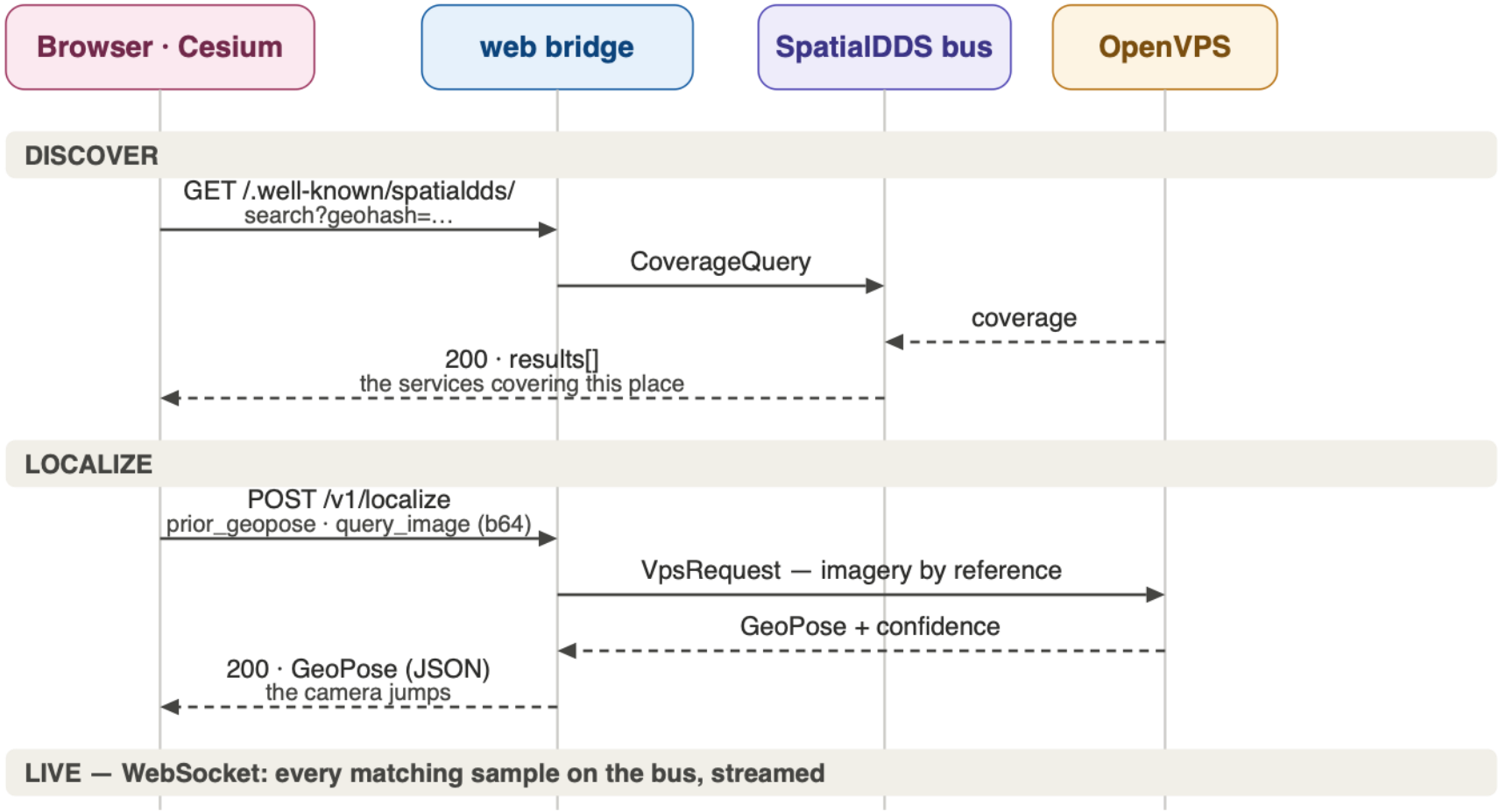
SpatialDDS + OpenVPS

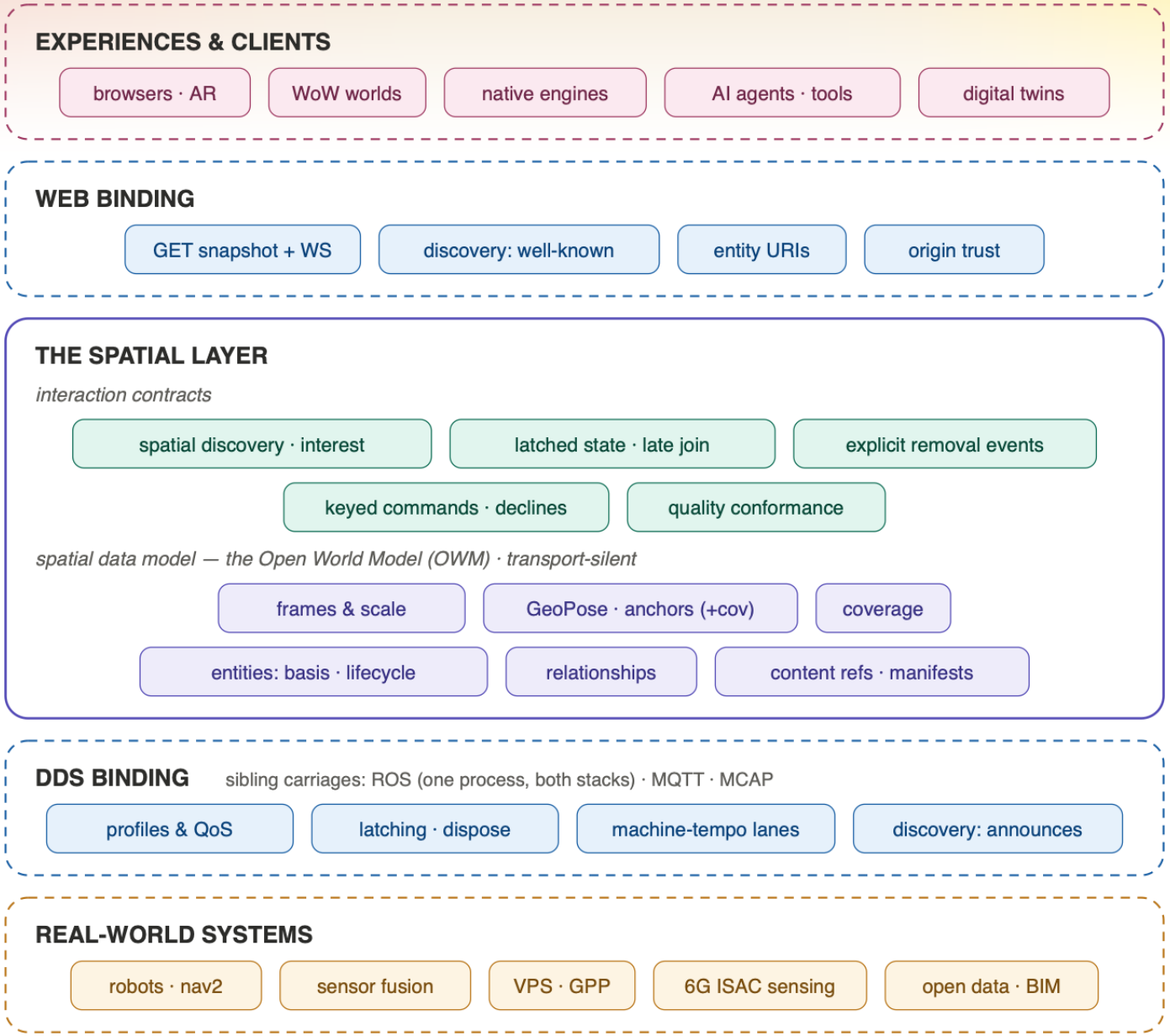
Device GeoPose via open spatial bus and visual positioning service





https://github.com/OpenArCloud/SpatialDDS-demo
https://github.com/OpenArCloud/openvps

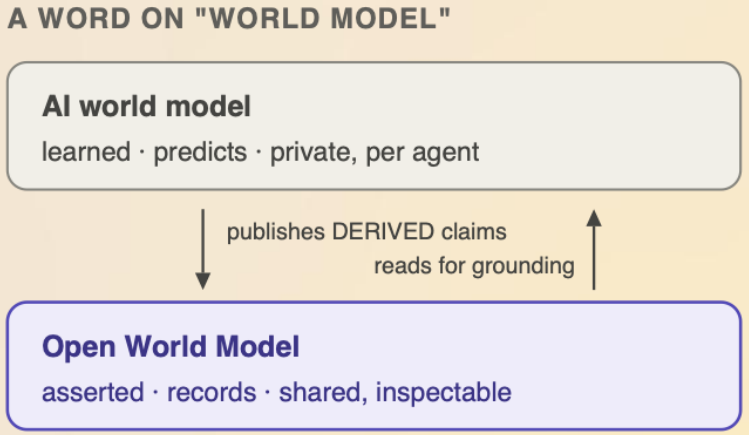

Browser · Cesium
web bridge
SpatialDDS bus
OpenVPS
DISCOVER
GET /.well-known/spatialdds/
search?geohash=…
CoverageQuery
coverage
200 · results[]
the services covering this place
LOCALIZE
POST /v1/localize
prior_geopose · query_image (b64)
VpsRequest — imagery by reference
GeoPose + confidence
200 · GeoPose (JSON)
the camera jumps
LIVE — WebSocket: every matching sample on the bus, streamed



https://spatialdds.org

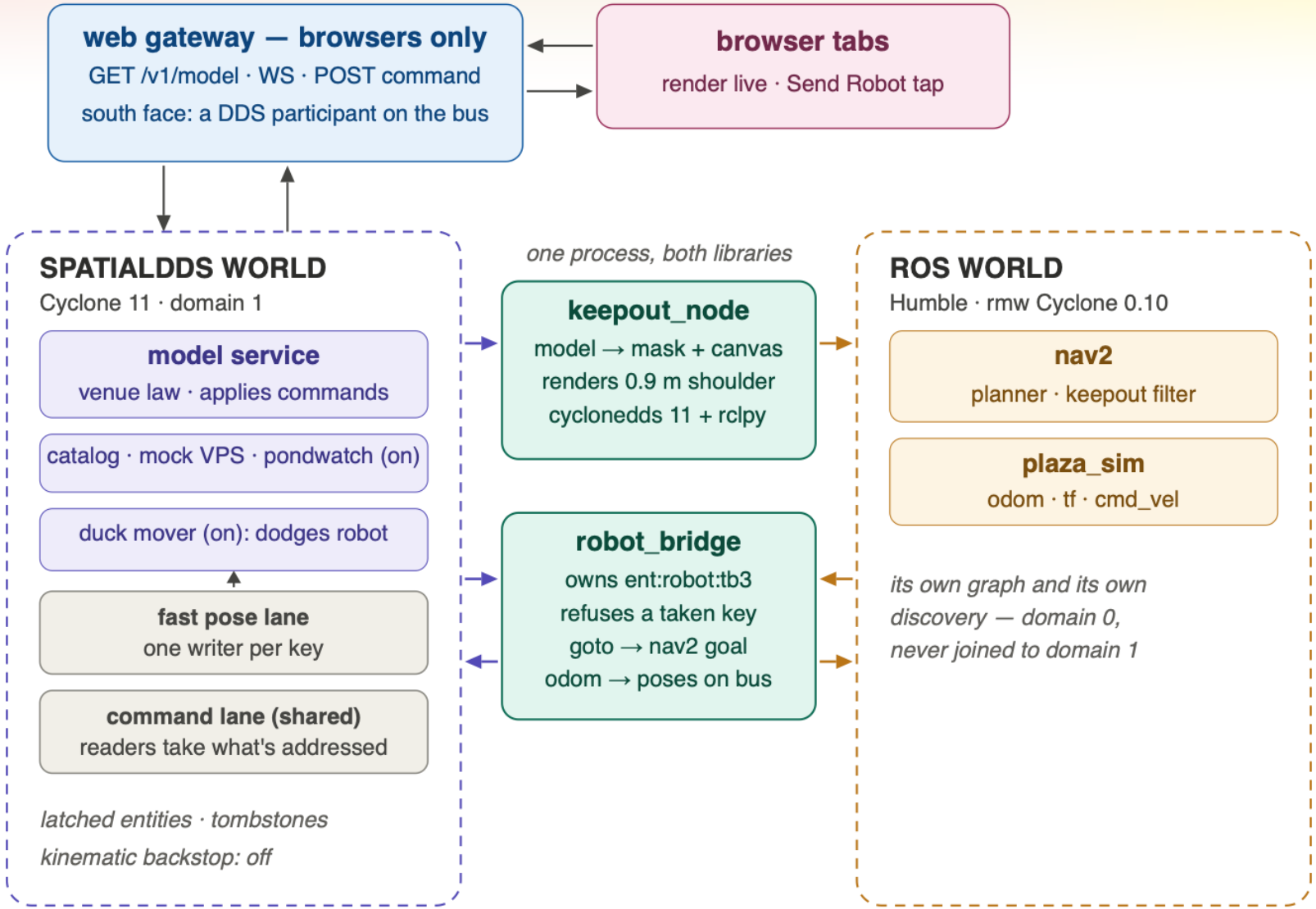
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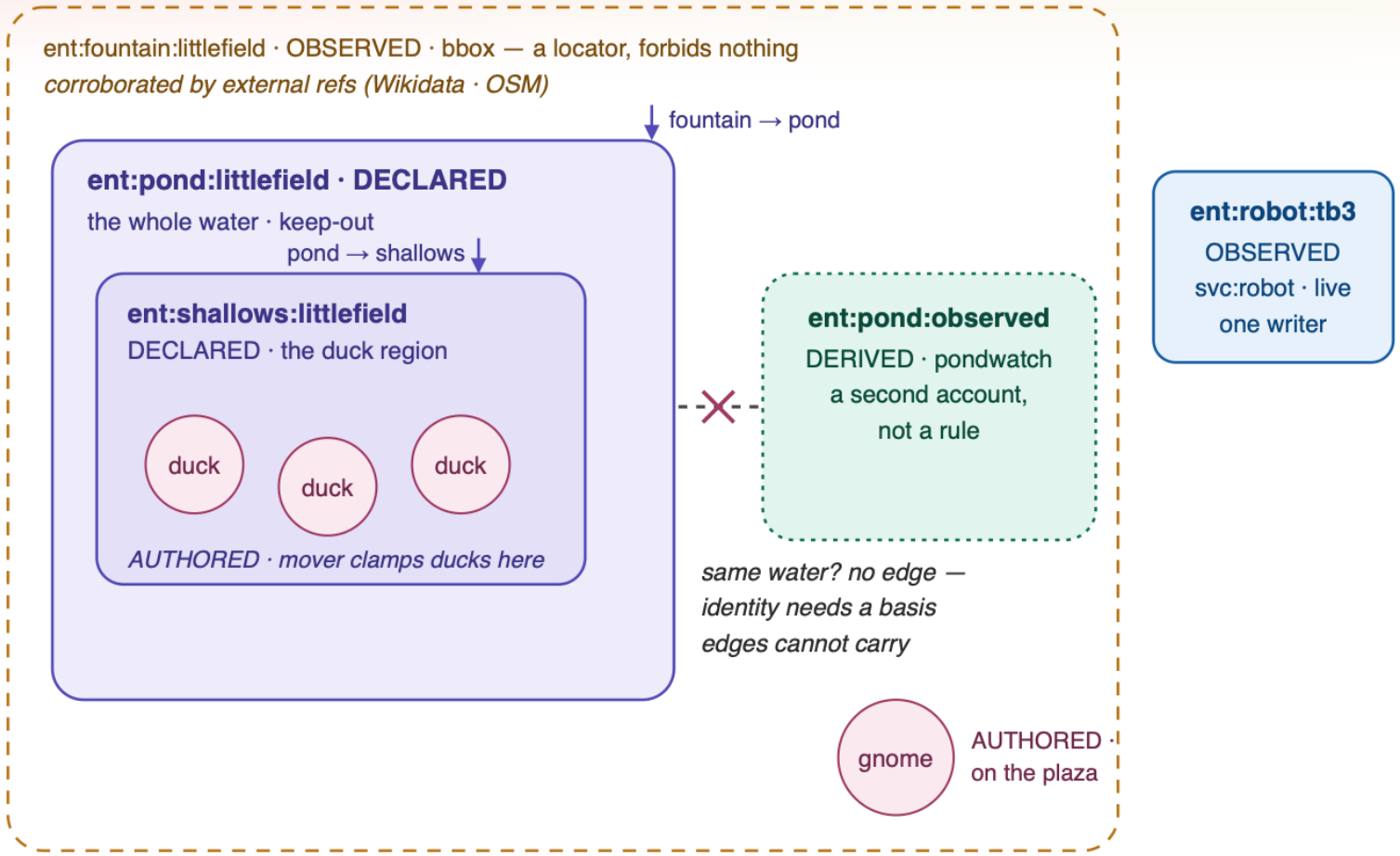
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Open World Model

SpatialDDS next steps


web gateway — browsers only
GET /v1/model · WS · POST command
south face: a DDS participant on the bus
browser tabs
render live · Send Robot tap
SPATIALDDS WORLD
Cyclone 11 · domain 1
model service
venue law · applies commands
catalog · mock VPS · pondwatch (on)
duck mover (on): dodges robot
fast pose lane
one writer per key
command lane (shared)
readers take what's addressed
latched entities · tombstones
kinematic backstop: off
one process, both libraries
keepout_node
model → mask + canvas
renders 0.9 m shoulder
cyclonedds 11 + rclpy
robot_bridge
owns ent:robot:tb3
refuses a taken key
goto → nav2 goal
odom → poses on bus
ROS WORLD
Humble · rmw Cyclone 0.10
nav2
planner · keepout filter
plaza_sim
odom · tf · cmd_vel
its own graph and its own
discovery — domain 0,
never joined to domain 1



A WORD ON "SCENE MODEL"

the tree a client renders
engines · web libraries · (WebXR: not one)

composed from

the shared model underneath
OWM · many parties · basis on claims

OBSERVED locates · DECLARED legislates · DERIVED opines · AUTHORED decorates

Thank you !

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